ECO702 Macroeconomics - Problem Set 2

1. Write the statement of the Contraction Mapping Theorem and prove it (again). While we covered this in lecture, it is important to be comfortable with the steps involved (i.e. using mathematical induction, the triangle inequality property of distance metrics, etc.).

2. Consider a neoclassical growth model with two sectors, one producing consumption goods and the other producing investment goods. Consumption is given by $C_t = F(K_{Ct}, L_{Ct})$ and investment is given by $I_t = G(K_{It}, L_{It})$, where K_{jt} is the amount of capital in sector j at the beginning of period t, and L_{jt} is the amount of labor used in sector j in period t. The total amount of labor in each period is equal to L (leisure is not valued by the household). Labor can be freely allocated in each period between the two sectors - $L = L_{Ct} + L_{It}$. Capital, by contrast, is sector-specific. Investment goods, however, can be used to augment the capital stock in either sector. In particular, the capital stock in sector j evolves according to

$$K_{j,t+1} = (1 - \delta)K_{jt} + I_{jt}$$

where $I_t = I_{Ct} + I_{It}$. A social planner seeks to maximize $\sum_{t=0}^{\infty} \beta^t u(C_t)$, given K_{C0} and K_{I0}, subject to the constraints on technology mentioned above. Note that although leisure is not valued in the utility function, the planner must nonetheless decide the allocation of L across the two sectors in each period.

   (a) Formulate the planner's optimization problem as a dynamic programming problem. What are the state variables? What are the control/choice variables? **Hint:** You should have two of each.

   (b) Find a set of first-order conditions and envelope conditions (using Benveniste-Scheinkman) that an optimal solution of the planning problem must satisfy.

   (c) Use your answer to part (b) to find a set of equations that determine the steady-state values of capital and labor (in each sector) in this economy.

   (d) Suppose that $F(K_{Ct}, L_{Ct}) = (K_{Ct})^{\alpha}(L_{Ct})^{1-\alpha}$ and $G(K_{It}, L_{It}) = (K_{It})^{\gamma}(L_{It})^{1-\gamma}$. Express the steady-state of the economy as a function of the parameters of the model.

3. Consider the following economic model (notation as in Gali's textbook). A representative household solves,

$$\max E_0 \sum_{t=0}^{\infty} \beta^t \Big(\frac{C_t^{1-\sigma} - 1}{1-\sigma} - \frac{N_t^{1+\psi}}{1+\psi} \Big)$$

subject to the sequence of budget constraints,

$$P_t C_t + Q_t B_t \leq B_{t-1} + W_t N_t + D_t$$

for $t = 0, 1, 2, ...$ Goods are produced by a representative firm with technology,

$$Y_t = A_t N_t^{1-\alpha}$$

where $a_t \equiv \ln A_t$ follows an exogenous process,

$$a_t = \rho_a a_{t-1} + \epsilon_t^a$$

Solve for equilibrium output and labor as functions of technology a_t. How does the labor response to a_t depend on σ?

4. Consider the same model as above, except that there is now a continuum of differentiated goods indexed by $i \in (0, 1)$ so that

$$C_t \equiv \Big(\int_0^1 C_{t,i}^{\frac{\epsilon-1}{\epsilon}} \Big)^{\frac{\epsilon}{\epsilon-1}}; \ \epsilon > 1$$

and,

$$C_{t,i} = Y_{t,i} = A_t N_{t,i}^{1-\alpha}$$

   (a) Derive the optimal demand for good i as a function of aggregate output and the relative price of good i.

   (b) Find the labor inputs used by firm i as a function of the relative price of good i and the aggregate price index P_t.

   (c) What is the optimal price set by firm i?

   (d) If prices are flexible and labor markets competitive, what is the dispersion in prices across different goods?

   (e) Now, assume that goods prices are sticky a la Calvo (1983). Show how the equilibrium allocations are inefficient even if there is a production subsidy that makes the steady state level of output efficient.